allowed. The related set theory concept is a subset. Roughly speaking, a proposition is true for all the elements in its corresponding subset and false for all the elements not in this subset. For example, if the set represents the possible states of a die, one can consider the proposition

A: The die shows an odd-numbered face.

The corresponding subset contains the three elements {1, 3, 5}.

Another proposition states

B: The die shows a number less than 4.

The corresponding subset contains the states {1, 2, 3}.

Every proposition has its opposite (also called its negation). For example,

not *A*: *The die does not show an odd-numbered face.*

The subset for this negated proposition is {2, 4, 6}.

There are rules for combining propositions into more complex propositions, the most important being **or, and,** and **not.** We just saw an example of **not,** which gets applied to a single subset or proposition. **And** is straightforward, and applies to a pair of propositions.[3] It says they are both true. Applied to two subsets, **and** gives the elements common to both, that is, the *intersection* of the two subsets. In the die example, the intersection of subsets *A* and *B* is the subset of elements that are both odd and less than 4. Fig. 1.5 uses a Venn diagram to show how this works.

The **or** rule is similar to **and**, but has one additional subtlety. In everyday speech, the word *or* is generally used in the exclusive sense—the exclusive version is true if one or the other of two propositions is true, but not both. However, Boolean logic uses the *inclusive* version of **or**, which is true if either or both of the propositions are true. Thus, according to the inclusive **or**, the proposition

Albert Einstein discovered relativity ***or*** *Isaac Newton was English*

is true. So is

Albert Einstein discovered relativity ***or*** *Isaac Newton was Russian.*